DE–8522 **11**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2014.

ORGANIC CHEMISTRY – I

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) State and explain the principle of microscopic reversibility. (4)

   (b) Explain steric inhibition of resonance with an example. (4)

   (c) Explain the stability of carbonium ion. (4)

   (d) Explain why C-C single bond length in propene is shorter than C-C single bond length in ethane. (2)

   (e) Fluoro ethanol is more acidic than ethanol-why? (2)

   (f) Explain the importance of Fulvenes. (4)

2. (a) Explain any two non-kinetic methods of determination of reaction mechanism. (6)

   (b) What are tropolones? How would you account for the aromatic character of tropolones? (6)

   (c) Explain S_N1 and S_{Ni} mechanism. (8)

3. (a) What are ambident nucleophiles? How will you prove that DMSO is an ambident nucleophile? (4)

(b) Discuss the mechanism of nitration of toluene. (4)

(c) Write brief note on partial rate factor. (4)

(d) Write note on phase transfer catalysis. (4)

(e) Discuss S_NAr mechanism. (4)

4. (a) Hydroboration is a regiospecific reaction – Explain. (6)

(b) Write an account on

(i) Addition of Grignard reagent to α, β - unsaturated carbonyl compound.

(ii) Darzen's reaction. (4+4)

(c) Compare elimination and substitution reaction. (6)

5. (a) Discuss with suitable examples, orientation of the double bond in elimination under various condition. (8)

(b) Discuss about the stereochemistry of the products formed Z-2-butene and E-2-butene $\xrightarrow{Br_2}$. (4)

(c) Write an account on :

(i) Michael addition

(ii) Mannich reaction. (4+4)

6. (a) Establish the structure of cyanidine chloride. (4)

(b) Explain how exhaustive methylation degradation method helps in determining the structure of alkaloids. (4)

(c) (i) Write the structure of the product formed when morphenol is fused with potassium hydroxide. (2)

(ii) Explain Zeisel method of estimation of $-OCH_3$ group. (2)

(d) Give the synthesis of oxazole. (4)

(e) How will you effect the following conversion? Norpinic acid from α – pinene. (4)

7. (a) Describe any two method for determining the size of the ring in sugars. (8)

(b) Write down the biosynthesis of terpenoids. (5)

(c) Discuss the structure of morphine. (7)

8. (a) How will you arrive the structure of cellobiose? (10)

(b) Explain the primary, secondary and tertiary structure of proteins. (10)

DE–8523 **12**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2014.

INORGANIC CHEMISTRY – I

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Draw Born - Haber cycle and explain the terms on it. (7)

   (b) Explain the formation of σ and π molecular orbitals. (6)

   (c) Discuss the presence of ionic character in a covalent bond. (7)

2. (a) State and explain HSAB principle with an example. (7)

   (b) Explain VSEPR theory with suitable examples. (7)

   (c) Bring out the difference between d^2sp^3 and sp^3d^2 hybridisation. (6)

3. (a) Draw and compare the structures of zincblende and wurzite. (8)

   (b) How is F - centre formed? Explain the effect of F centre on properties of solids. (6)

   (c) Explain the consequences of metal excess defect. (6)

4. (a) Give the conditions for the formation of schottky and frenkel defects. (5)

(b) Explain briefly the structures of cadmium iodide and nickel arsenide. (9)

(c) Write short note on - stripping and pick - up reaction. (6)

5. (a) Neutron activation analysis is a non-destructive analysis. Justify. (6)

(b) Discuss the application of liquid drop model with respect to the mechanism of nuclear fission. (8)

(c) In nuclear reactor both 235_U and 238_U are used. Why? (6)

6. (a) Give the principle and methodology of ion exchange method for the separation of individual Lanthanides. (8)

(b) Explain the spectral and magnetic properties of actinides. (8)

(c) Discuss the use of lanthanides as shift reagents. (4)

7. (a) Define lanthanide contraction. Explain the consequences of lanthanide contraction. (10)

(b) The unit cell of TiO_2 is not a perfect cubic – Explain. (4)

(c) Define and explain the terms with an example – Isopoly and heteropoly acids. (6)

8. (a) Explain various structural units of silicates with examples. (9)

(b) Comment on the structure of clay minerals. (6)

(c) Discuss the Keggin's structure of hetero – Polyacids. (5)

DE–8524 **13**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2014

PHYSICAL CHEMISTRY – I

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) What do you mean by the term "Entropy of Mixing"? Explain. (6)

   (b) Derive Maxwell relations using thermodynamic equation of states. (7)

   (c) Derive Gibbs-Helmholtz equation. Explain the criteria for spontaneity and conditions of equilibrium with the aid of Gibb's free energy function. (7)

2. (a) What do you mean by the term Fugacity? How will you determine fugacity of a real gas? (7)

   (b) Derive Butler-Volmer equation and explain its significance. (6)

   (c) Write notes on Debye Huckel limiting law. (7)

3. (a) Derive Debye-Huckel Onsager equation and explain about its validity and modifications. (8)

(b) The wavelength of a moving bullet of mass 10 mg is 3.31×10^{-27} m. Calculate its kinetic energy ($h = 6.626 \times 10^{-34}$ Js). (5)

(c) Derive the Schrodinger wave equation. (7)

4. (a) Derive the expression for studying the particle constrained to move is an one dimensional box. (8)

(b) Discuss about different (any three) types of electrodes with their respective cell reactions. (6)

(c) Give each one example for parallel, consecutive and opposing reactions. (6)

5. (a) Describe the Lindemann hypothesis for the study of kinetics of unimolecular reactions. Explain the merits and demerits of the hypothesis. (8)

(b) What are fast reactions? How are they studied with the help of relaxation techniques? (7)

(c) What do you understand by the term kinetic isotopic effect? Explain. (5)

6. (a) State and explain about the principles of microscopic reversibility. (6)

(b) Discuss the kinetics – mechanism and quantum yield of photochemical combination of $H_2 - Cl_2$ reaction. (8)

(c) Explain the terms photosensitisation and chemiluminescence with proper examples. (6)

7. (a) Describe the various photophysical processes with the help of Jabalonski diagram. (10)

(b) What do you mean by steady state approximation? Explain with the help of an example. (5)

(c) Write notes on potential energy surfaces. (5)

8. (a) Discuss any three principles for the conversion and storage of solar energy with proper illustrations. (8)

(b) Discuss about the different types of potentiometric titrations. (6)

(c) Explain the term "degeneracy" with the help of the solutions of the wave equation of three dimensional box. (6)

DE–8525 **14**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2014

INSTRUMENTAL METHODS OF ANALYSIS

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks

$(5 \times 20 = 100)$

1. (a) Give the effect of substituents and solvents in the position and intensity of absorption bands? (8)

   (b) How will you determine the formation of charge transfer complexes using UV–Visible spectroscopy? (7)

   (c) Define and explain Hook's law. (5)

2. (a) Explain the following :

   (i) Overtone. (4 + 4 + 4)

   (ii) Combination bands.

   (iii) Fermi resonance.

   (b) Give the applications of IR-Spectra in the study of hydrogen bonding? (8)

3. (a) Give the detail study of chemical shift and factors affecting chemical shift. (10)

   (b) Write a note on Nuclear Overhauser effect. (5)

   (c) Write the principle of ESR spectra. (5)

4. (a) Compare the NMR and ESR spectral techniques. (8)

   (b) Hyper–fine splitting in ESR spectra. (7)

   (c) What is meant by Double resonance technique? (5)

5. (a) Explain the following :

   (i) Meta stable ion. (4)

   (ii) Isotopic peaks. (4)

   (iii) Nitrogen rule. (5)

   (b) Discuss the melafferty rearrangement with an example. (7)

6. (a) Explain α–halo ketone rule with an example. (7)

   (b) Discuss briefly the working of HPLC? (7)

   (c) What is meant by thermogram? Explain with an example. (6)

7. (a) Write note on :

   (i) Ion selective electrodes. (6)

   (ii) Amperometry titrations. (6)

   (b) Give the principle and applications of cyclic voltammetry? (8)

8. (a) Define the term accuracy and precision. (6)

   (b) Discuss the rejection of a result on the basis of Q-test and t-test. (8)

   (c) Draw a polarogram and explain the various parameters in it. (6)

DE–8526 **21**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2014.

ORGANIC CHEMISTRY — II

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carriy 20 marks.

($5 \times 20 = 100$)

1. (a) Discuss the mechanism of Demzonev rearrangement and stern and Favorski rearrangement. (12)

   (b) Explain the following reactions with suitable example.

   (i) Lead tetraacetate oxidation

   (ii) Wolff – kishner reduction. (8)

2. (a) Discuss Chan-Ingold-Prelog rules for assigning the configuration of a Chairal Carbon. (8)

   (b) Write the rule for writting E, Z- nomenclature for carbon – carbon double bonded compounds. (6)

   (c) What is atropisomerism? Explain it with suitable examples. (6)

3. (a) Discuss any four reactions of free radicals. (8)

   (b) Discuss elaborately about the conformation of mono substituted and disubstituted cyclohexane. (12)

4. (a) Write note on sensitization and quenching. (8)

   (b) Discuss elaborately about frontier molecular orbital approach. (12)

5. (a) Elucidate the structure of vitamin A. (12)

   (b) Discuss the biosynthetic route for cholesterol. (8)

6. (a) Explain retro synthetic analysis of simple organic compound with suitable example. (8)

   (b) Explain Umpolung synthesis with example. (8)

   (c) Discuss briefly about the stereospecific reaction with suitable example. (4)

7. (a) Explain the following reactions with examples.

      (i) Gomberg

      (ii) Sandmayer

      (iii) Barton. (9)

   (b) Give detail account on quantum efficiency. (8)

   (c) Explain the primary process in the Norrish Type I. (3)

8. (a) Establish the structure of progesterone. (8)

   (b) Elucidate the structure of Vitamin D. (8)

   (c) Write the mechanism of Birch reduction. (4)

DE–8527 **22**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2014.

INORGANIC CHEMISTRY – II

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries 20 marks.

$(5 \times 20 = 100)$

1. (a) $[CO(NH_3)_6]^{3+}$ is a low spin complex while $[COF_6]^{3-}$ is a high spin complex. Explain. (6)

   (b) Draw the crystal field splitting energy level diagram of $(Ni(CN)_4)^{2-}$ and account for its diamagnetic character. (6)

   (c) Discuss the splitting and energy level diagram of d-orbitals of Cu^{2+} ion in an octahedral complex caused by Jahn–Teller effect. (8)

2. (a) Providing suitable example, distinguish between inner and outer sphere mechanisms for electron transfer. (8)

   (b) What is meant by trans effect? What are its applications? (6)

   (c) Discuss the polarographic method for the determination of stability constant for a complex species. (6)

3. (a) Explain the spectra $[Ti(H_2O)_6]^{3+}$ and $[Cu(H_2O)_6]^{2+}$ with orgel diagram. (8)

(b) Discuss the steps involved in the evaluation of Dq and β. Illustrate with an example. (6)

(c) Describe the Guoy balance with a diagram for the measurement of magnetic susceptibility. (6)

4. (a) Explain briefly an Wacker process. (6)

(b) Illustrate the main catalytic cycle in the homogeneous hydrogenation of an alkene by rhodium–phosphine complex. (8)

(c) How is Zeise's salt prepared? Discuss its structure and bonding. (6)

5. (a) How are the following chelating agents used therapeutically to treat the problem caused by the presence of toxic element?

(i) Cis–diamminedichloroplatinum(II) and

(ii) D-Penicillamine. (6)

(b) Discuss the function of chlorophyll in photosynthesis. (6)

(c) Discuss the important structural aspects and functions of hemoglobin and myoglobin. (8)

6. (a) Mn_3O_4 is a normal spinel but Fe_3O_4 is an inverse spinel. Explain. (6)

(b) List the various factors influencing the magnitude of crystal field splitting. (6)

(c) Explain the terms isomerization and racemisation. (8)

7. (a) What are Tanabe-Sugano diagrams? How are they useful to interpret electronic spectrum of transition metal complexes? (8)

(b) What is fluxional behaviour of molecules? Explain it with examples. (5)

(c) Explain the catalytic mechanism of Zeigler – Natta catalyst for the polymerisation. (7)

8. (a) Explain the terms oxidative addition and reductive elimination with examples. (4)

(b) Discuss briefly about Bio-redox reagents. (6)

(c) Explain the mechanism involved in the ion-transport across membrane. (6)

(d) Explain how ferredoxins differ from rubredoxins. (4)

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DE–8528 **23**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2014.

PHYSICAL CHEMISTRY – II

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Give the comparison of three statistical distribution laws. (12)

   (b) Derive an expression for the translational partition function. (8)

2. (a) Discuss the application of variation method in solving hydrogen atom problem. (14)

   (b) Write note on Pauli exclusion principle. (6)

3. (a) Find out the symmetry of normal modes of vibration of water molecule. Predict the IR and Raman active fundamental vibrations of water molecule. (10)

   Character table of C_2V point group.

C_2 v	E	C_2	$\sigma_{v(xz)}$	$\sigma_{v(yz)}$		
A_1	1	1	1	1	Z	x^2, y^2, z^2
A_2	1	1	–1	–1	R_x	xy
B_1	1	–1	1	–1	X, R_y	xz
B_2	1	–1	–1	1	Y, R_z	yz

   (b) Give the matrix representations of E, C_n, i and σ_{xy}. (10)

4. (a) Derive Bronsted – Bjerrum equation. (10)

(b) Discuss the effect of pH and temperature on enzyme catalysed reaction. (10)

5. (a) What are the different types of physical adsorption isotherm? (8)

(b) Bring out the difference between physisorption and chemisorption. (8)

(c) How would you test the validity of freundlich adsorption isotherm? (4)

6. (a) Write note on Onsager reciprocal relations. (6)

(b) Discuss the application of HMO method to butadiene. (6)

(c) What are point groups? Discuss the systematic assignment of point groups to molecules. (2+6)

7. (a) Derive an expression for Fermi–Dirac statistics. (6)

(b) Write note on slater determinant. (6)

(c) Derive Michaelis – menton equation. (8)

8. (a) Write note on reducible and irreducible representation. (4)

(b) Briefly discuss about the secondary salt effect on the rate of the reaction. (4)

(c) Discuss the applications of adsorption phenomenon. (12)

DE–8529 **24**

DISTANCE EDUCATION

M.Sc.(Chemistry) DEGREE EXAMINATION, MAY 2014.

APPLIED CHEMISTRY

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) How will be waste water, treated with help of biological methods? (7)

   (b) What are radioactive wastes? How are they disposed? (5)

   (c) What is green house effect? How is green house effect produced? What are the major sources of green house effect gases? Explain the consequences of green house effect. (8)

2. (a) Describe the chief process involved in the conversion of polluted river water coming from an industrial town into water fit for domestic use. (7)

   (b) What is acid rain? Explain the theory of acid rain.(7)

   (c) Indicate those activities of man that have resulted in air pollution. (6)

3. (a) What is meant by corrosion of metals? How does it occur and how can it he exhibited? (8)

   (b) What is potential/pH diagram? Draw and explain the diagram for Fe/H_2O system. (8)

   (c) Give an account of electroplating of nickel. (4)

4. (a) Which metal is used in the cathodic protection of corrosion? Explain the methods and advantages of it. (7)

(b) Illustrate the principles and applications of cladding. (7)

(c) How will you do electro plating using simple and complex salts? Explain it with suitable examples. (6)

5. (a) What are the solution and bulk polymerizations? Explain the techniques briefly. (8)

(b) What is initiator? Discuss its types and efficiency with suitable examples. (7)

(c) Write note on biodegradable polymers. (5)

6. (a) How can nanomaterials be characterised by TGA, DTA and DSC? (12)

(b) Explain the synthesis of nano materials using sol–gel method. (5)

(c) Write note on sputtering. (3)

7. (a) What are SEM and TEM? How it could be used for the characterization of nanoparticles? (10)

(b) Write note on fire retardant. (5)

(c) What is degree of polymerization? Explain it with help of suitable examples. (5)

8. (a) Write basic program for the calculation of.

(i) pH

(ii) Solubility product (6+6)

(b) What are the terms used in email search engines? Explain each with examples. (8)